Customer Behavior Analysis

End-to-end analysis of 3.9K customers' shopping habits, revealing actionable insights into demographics, category performance, and purchasing behaviors.

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Executive Summary

3.9K

Customers Analyzed

Comprehensive dataset

$59.76

Average Purchase

Per transaction amount

3.75

Average Rating

Customer review score

End-to-end analytical workflow using Python for data cleaning, SQL Server for exploratory analysis, and Power BI for visualization.

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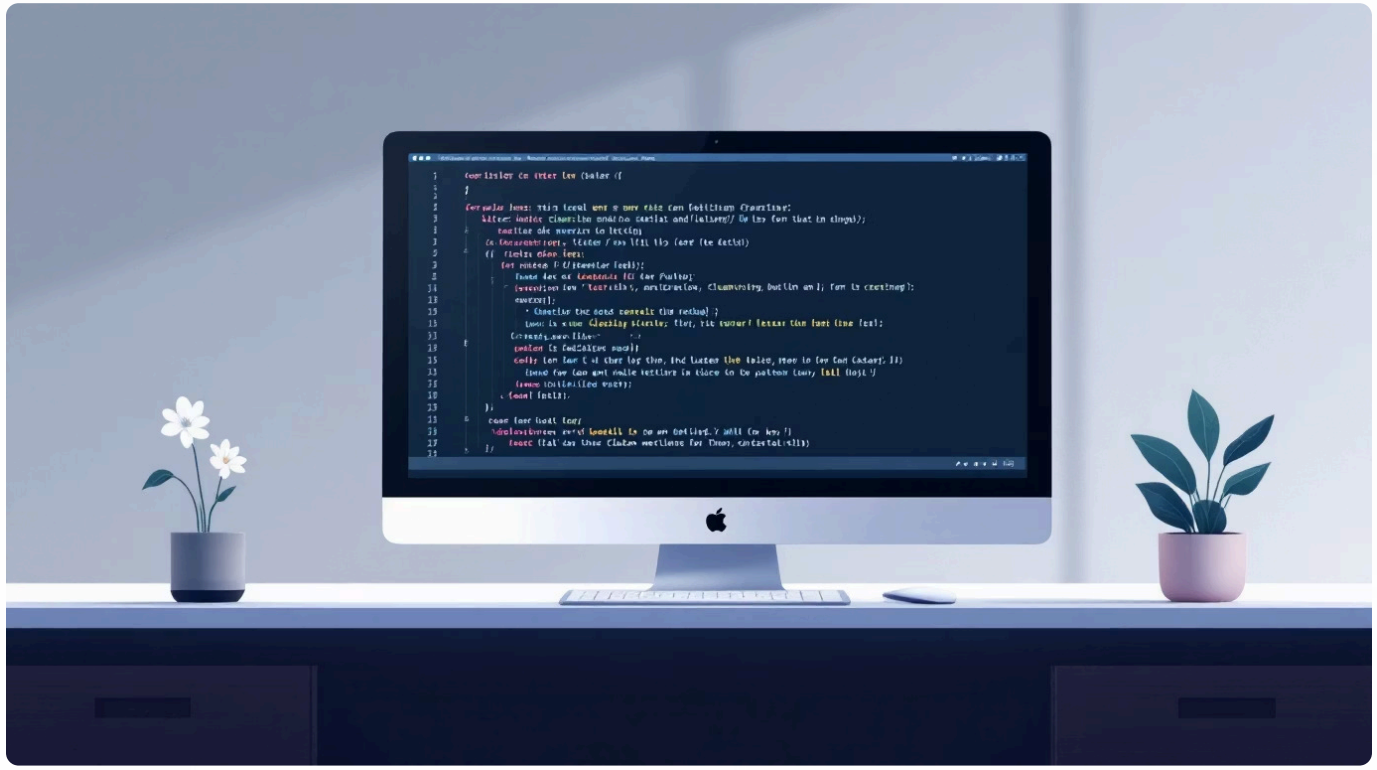
Data Preparation & Cleaning

Python Processing

Raw dataset processed using Pandas library for analysis readiness.

- Column standardization for SQL compatibility
- Missing data filled with median ratings
- Feature engineering for age groups



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Feature Engineering Highlights

Age Groups

Customers segmented into four quartiles: young adult, adult, middle aged, and senior.

Purchase Frequency

Categorical frequencies mapped to numeric days (e.g., Weekly = 7, Annually = 365).

Database Integration

Cleaned data pushed to SQL Server using SQLAlchemy.

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Exploratory Data Analysis

01

Revenue Demographics

Total revenue compared by gender and age group.

02

Discount & Spending

Identified customers using discounts while spending above average.

03

Product Performance

Top products ranked by review ratings and purchase frequency.

04

Customer Segmentation

Segmented into new, returning, and loyal customer groups.

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Dashboard Overview

Interactive Power BI dashboard with filters for subscription status, gender, category, and shipping type.

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Key Insights

Clothing Dominates

Category generates over $100K in revenue and 1,750 sales, vastly outperforming Accessories, Footwear, and Outerwear.

Young & Middle-Aged Lead

These age groups are top performers for both sales volume and revenue generation.

Subscription Gap

73% of customers are unsubscribed, representing massive recurring revenue potential.

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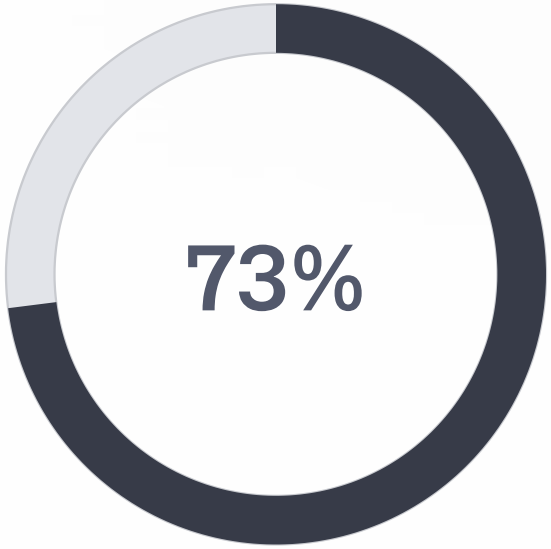
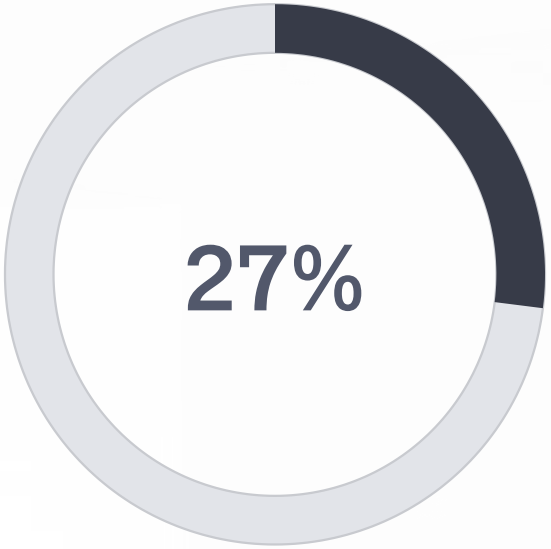
Customer Segmentation



Unsubscribed

3.9K total customers



Subscribed

Current subscription rate

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Recommendations

Targeted Subscription Campaigns

Launch campaigns aimed at loyal customers (>10 purchases) to convert them into subscribers.

Inventory & Marketing Focus

Allocate more resources to Clothing category, targeting young adult and middle-aged demographics.

Leverage Discounts

Build strategic bundle offers using high-discount products to encourage spending above $59.76 average.

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Next Steps

Implement Campaigns

Execute targeted marketing to loyal customer segments.

Monitor Performance

Track subscription conversion rates and revenue impact.

Optimize Strategy

Refine approaches based on ongoing data analysis.

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